Extreme Low-fat Diet as a Synergistic Risk Factor for Gastric Diseases: Resolving the Japanese Diet Paradox

ZHU, GUANGWEN[a,*]

[a] *Harbin Institute of Technology, Harbin, China*

Abstract

The traditional Japanese dietary pattern, characterized by extremely low fat intake (<15% of energy), has long been regarded as a paradigm of healthy eating. Paradoxically, Japan remains one of the developed countries with the highest gastric cancer incidence globally. Current explanations focusing on high salt intake and *Helicobacter pylori* infection fail to fully account for this elevated risk.

We propose a novel hypothesis: long-term extreme low-fat diet is an independent synergistic risk factor for gastric diseases in Japan. Chronic inadequate dietary fat intake impairs gastrointestinal mucosal barrier integrity through multiple mechanisms—reduced phospholipid synthesis, diminished prostaglandin E2 production, and disrupted gut microbiota homeostasis. This compromised mucosal barrier creates a "vulnerable state" that amplifies the damaging effects of established risk factors such as high salt consumption and *H. pylori* infection.

Supporting evidence includes: (1) animal studies demonstrating that essential fatty acid deficiency impairs gastric mucosal protection and increases susceptibility to injury; (2) the Japan Collaborative Cohort Study showing a significant inverse association between dietary fat intake and gastric cancer risk (HR 0.66 for highest vs. lowest quintile); (3) temporal trends showing that increased fat consumption in Japan over 50 years correlates with declining gastric cancer mortality.

*Corresponding author.
Email address: `ares.z@hotmail.com` (ZHU, GUANGWEN)

This hypothesis provides a mechanistic explanation for the Japanese diet paradox and suggests that dietary fat optimization (20–30% of energy) should be considered as a complementary gastric cancer prevention strategy.
Keywords: Extreme low-fat diet, Gastrointestinal mucosal barrier, Gastric cancer, Japanese dietary pattern, Synergistic risk factor, Prostaglandin E2

---

1. Introduction

According to GLOBOCAN 2020, Japan's age-standardized gastric cancer incidence rate reaches 27.5 per 100,000, ranking first among developed countries—far exceeding rates in Western nations (United States: 5.6/100,000; Germany: 7.8/100,000) [1]. This elevated risk persists despite Japan's advanced healthcare system and globally admired dietary culture.

Concurrently, the traditional Japanese diet has been extensively promoted as a model of healthy eating, associated with the world's highest life expectancy [2]. A core characteristic of this dietary pattern is extremely low fat intake: traditional Japanese residents consumed less than 10g of cooking oil daily, with fat providing less than 15% of total energy—substantially below the 20–30% range recommended by most international dietary guidelines [3, 4].

This creates a striking epidemiological contradiction: a population following an ostensibly "healthy low-fat diet" experiences the highest gastric cancer rates among developed nations.

1.1. Limitations of Current Explanations

The prevailing explanations for Japan's high gastric cancer incidence focus on high salt intake [5] and *Helicobacter pylori* infection [6]. However, these explanations face significant challenges:

1. **Declining risk factors, persistent high incidence:** Japan's *H. pylori* eradication rate has exceeded 80% following national screening programs, and salt intake has decreased by over 30% since the 1960s [7, 8]. Yet gastric

cancer incidence remains substantially elevated compared to Western countries.

2. **The Asian enigma:** Other Asian populations with high *H. pylori* prevalence (e.g., India, Bangladesh) have remarkably low gastric cancer rates [9].
3. **Residual risk:** Despite declining prevalence of known risk factors, gastric cancer incidence in Japan remains higher than in Western countries, suggesting the involvement of additional, underrecognized factors [10].

These observations suggest the existence of underrecognized contributing factors.

2. The Hypothesis

We propose the following hypothesis:

> *Long-term extreme low-fat dietary intake (<15% of energy from fat) is an independent synergistic risk factor for gastric diseases. Chronic insufficient dietary fat impairs gastrointestinal mucosal barrier integrity and diminishes mucosal defense and repair capacity, thereby amplifying the damaging effects of established risk factors (high salt, H. pylori infection) and contributing to elevated gastric disease incidence.*

This hypothesis does not suggest that low-fat diet directly causes gastric cancer. Rather, it proposes that extreme low-fat intake creates a state of mucosal vulnerability that magnifies the effects of other carcinogenic factors—a synergistic relationship that explains why Japanese populations experience higher gastric cancer rates despite similar or lower exposure to individual risk factors compared to some other populations.

3. Evaluation of the Hypothesis

3.1. Mechanistic Basis: Dietary Fat and Mucosal Barrier Integrity

The gastrointestinal mucosal barrier comprises multiple defensive layers: the mucus-bicarbonate layer, epithelial cell layer with tight junctions, and underlying lamina propria [11]. Dietary fat contributes to barrier integrity through several interconnected mechanisms.

3.1.1. Phospholipid Membrane and Mucus Layer

Essential fatty acids—linoleic acid and α-linolenic acid—are obligate components of cell membrane phospholipids. Gastric epithelial cells have high turnover rates (complete renewal every 3–5 days), requiring continuous phospholipid supply [12].

Phosphatidylcholine, the predominant phospholipid in gastrointestinal mucus, forms a hydrophobic barrier on the mucosal surface that resists acid and pepsin penetration [13, 14]. Animal studies demonstrate that dietary fat composition significantly modulates intestinal tight junction integrity and epithelial permeability [15], while essential fatty acid deficiency impairs mucosal barrier function [16].

3.1.2. Prostaglandin-Mediated Protection

Prostaglandin E2 (PGE2) is a central mediator of gastric mucosal defense, stimulating mucus and bicarbonate secretion, enhancing mucosal blood flow, and promoting epithelial repair [17]. PGE2 synthesis requires arachidonic acid, which derives from dietary linoleic acid [18].

Konturek et al. demonstrated that dietary fat restriction significantly reduces gastric mucosal PGE2 levels, with corresponding decreases in mucus secretion and mucosal blood flow [19]. Animal studies have consistently shown that essential fatty acid deficiency impairs prostaglandin-dependent gastroprotection [20].

3.1.3. Gut Microbiota-Mucosal Barrier Axis

A systematic review synthesizing 44 human dietary intervention studies found that extreme macronutrient restriction, including very low-fat diets, consistently reduces gut microbial diversity [21]. Specifically, very low-fat diets decrease the abundance of beneficial bacteria such as *Faecalibacterium prausnitzii* and *Akkermansia muciniphila*, which are essential for short-chain fatty acid (SCFA) production and mucus layer maintenance.

SCFAs, particularly butyrate, upregulate tight junction protein expression and stimulate mucin secretion [22, 23]. Reduced SCFA production compromises epithelial barrier function throughout the gastrointestinal tract.

3.2. Experimental and Biomarker Evidence

Multiple experimental studies support the hypothesis:

Essential fatty acid deficiency models: Hollander and Tarnawski demonstrated that rats fed essential fatty acid-deficient diets develop spontaneous gastric erosions, with mucosal PGE2 levels reduced by 60–70% and delayed healing [16].

Dietary linoleic acid studies: Schepp et al. demonstrated that dietary linoleic acid modulates gastric mucosal PGE2 release and protects against stress-induced mucosal damage in rats [20]. Diets deficient in essential fatty acids resulted in impaired gastroprotection, which was reversed by linoleic acid supplementation.

Synergistic dietary effects: The concept of dietary factor interactions in gastric mucosal injury is further supported by population-level biomarker studies. Tsugane et al. demonstrated that biological markers of stomach cancer risk are influenced by multiple dietary factors acting in combination [24], consistent with the hypothesis that inadequate fat intake may exacerbate injury from other dietary aggressors such as high salt.

3.3. Epidemiological Evidence: Japanese Cohort Studies

Large-scale Japanese cohort studies provide robust epidemiological evidence.

The Japan Collaborative Cohort (JACC) Study analyzed data from 42,042 Japanese men followed for 14 years [25]:

- Dietary fat intake was significantly inversely associated with gastric cancer incidence
- Compared to the lowest quintile of fat intake (<15% of energy), the highest quintile (>25%) had 34% lower gastric cancer risk (HR 0.66, 95% CI 0.47–0.92)
- The association remained significant after adjustment for salt intake, *H. pylori* serology, smoking, and alcohol consumption

Additionally, the protective role of n-3 polyunsaturated fatty acids against gastrointestinal cancers has been documented in Asian populations [18], with mechanistic evidence suggesting that these fatty acids may attenuate *H. pylori*-related mucosal inflammation [17].

3.4. Temporal Trends

Ecological analyses of temporal trends in Japan provide additional support [10, 26]:

- Mean daily dietary fat intake increased from ∼10g to ∼55g between 1950–2010
- Fat as percentage of energy increased from ∼8% to ∼27%
- Age-standardized gastric cancer mortality decreased from 80.7 to 24.8 per 100,000 in men

During the same period, salt intake decreased by only ∼30%. Multivariate modeling suggested that increased fat intake was an independent predictor of declining gastric cancer mortality [26].

Table 1: Dietary fat intake and gastric cancer rates in developed countries

Country	Fat (% energy)	Gastric cancer ASR (/100,000)
Japan	26.8	27.5
South Korea	22.4	32.1
United States	36.2	5.6
France	40.1	5.8
Germany	37.8	7.8

3.5. International Comparative Evidence

Cross-national comparisons support dietary fat's role in gastric cancer epidemiology [27]:

These ecological data suggest an inverse relationship between national dietary fat intake and gastric cancer rates, though such cross-national comparisons must be interpreted cautiously due to potential confounding factors.

Migration studies of Japanese-Americans in Hawaii showed that gastric cancer rates declined across generations as dietary fat intake increased with acculturation [28].

3.6. The Synergistic Mechanism

The hypothesis proposes that extreme low-fat diet amplifies traditional risk factors rather than acting independently (Figure 1).

Amplification of high-salt injury: With thinner mucus layer and reduced phospholipid barrier, salt penetrates more rapidly to the epithelial surface. PGE2-dependent repair mechanisms are compromised, so salt-induced injuries persist longer. The biological plausibility of this synergistic effect is supported by the established roles of both dietary fat (in mucosal defense) and salt (in mucosal injury) [5, 17].

Enhanced *H. pylori* pathogenicity: Compromised mucosal barriers facilitate *H. pylori* colonization and virulence factor delivery. The intact mucus layer and phospholipid barrier normally impede bacterial adhesion to epithelial

cells [14]. When these defenses are weakened by inadequate dietary fat, *H. pylori* may colonize more effectively and exert greater pathogenic effects.

Cumulative damage: The prolonged course of gastric carcinogenesis (20–30 years) allows small differences in mucosal protection to compound over time, significantly increasing progression through the atrophic gastritis → intestinal metaplasia → dysplasia → carcinoma sequence [29].

Proposed Mechanistic Model

Extreme Low-fat Diet (<15% energy)

↓

↓ Phospholipid synthesis | ↓ PGE2 synthesis | ↓ Gut microbiota diversity

↓

↓ Mucus layer | ↓ Tight junctions | ↓ SCFA production

↓

Compromised Mucosal Barrier ("Vulnerable State")

↓

Amplified damage from: High salt | *H. pylori* | Stress/smoking/alcohol

↓

Accelerated progression: Normal → Gastritis → Atrophy → Metaplasia → Cancer

Figure 1: Proposed mechanistic model for synergistic effects of extreme low-fat diet on gastric carcinogenesis.

4. Consequences of the Hypothesis and Discussion

4.1. Resolving the Japanese Diet Paradox

The hypothesis resolves the apparent contradiction between Japan's "healthy" low-fat diet and high gastric cancer rates by recognizing that:

1. The traditional Japanese diet is not uniformly healthy; its extreme low-fat characteristic creates a specific vulnerability
2. The same dietary pattern carries both benefits (low saturated fat, high fish) and risks (extreme low total fat, high salt)
3. Net health outcomes depend on the balance of these factors and their interactions

Modern Japanese dietary patterns, with increased fat intake while maintaining other healthy characteristics, may represent an optimization that preserves cardiovascular benefits while reducing gastric cancer risk.

4.2. Implications for Dietary Guidelines

Current dietary guidelines appropriately emphasize reducing excessive fat intake for cardiovascular health. However, our hypothesis suggests guidelines should also caution against extreme fat restriction:

- **Recommended range:** Fat intake of 20–30% of total energy balances cardiovascular and gastrointestinal health
- **Minimum threshold:** Fat intake below 15% of energy for extended periods may carry gastrointestinal health risks
- **Fat quality:** Emphasis on unsaturated fats provides cardiovascular benefits while maintaining mucosal protection

4.3. Implications for Gastric Cancer Prevention

Current gastric cancer prevention focuses on *H. pylori* eradication, salt reduction, and smoking cessation. Our hypothesis suggests adding dietary fat optimization as a complementary strategy, particularly for individuals with other risk factors.

Clinicians counseling patients with or at risk for gastric diseases should:

- Assess dietary fat intake as part of nutritional evaluation

- Avoid recommending extreme low-fat diets for patients with chronic gastritis
- Consider n-3 PUFA supplementation in patients with inadequate intake

4.4. Limitations and Future Research

The evidence reviewed is predominantly observational, precluding definitive causal inference. Key limitations include:

- Lack of randomized controlled trials with gastric endpoints
- Research paradigm bias (low-fat diets historically assumed healthy)
- Potential residual confounding in epidemiological studies
- Most evidence from Japanese populations

Future research priorities include:

1. **Mechanistic studies:** Human mucosal biopsy studies comparing barrier function between extreme low-fat vs. adequate fat consumers
2. **Mendelian randomization:** Using genetic variants associated with fat metabolism as instrumental variables
3. **Interaction analyses:** Systematic examination of fat × salt and fat × *H. pylori* interactions in existing cohorts
4. **Intervention trials:** Dietary fat optimization in individuals with atrophic gastritis, with histological regression as endpoint

5. Conclusions

We propose that long-term extreme low-fat diet serves as a synergistic risk factor for gastric diseases by impairing mucosal barrier integrity and amplifying the effects of established risk factors. This hypothesis is supported by:

1. Mechanistic evidence linking dietary fat to mucosal barrier function through phospholipid synthesis, PGE2 production, and gut microbiota

2. Experimental evidence demonstrating increased mucosal vulnerability with fat restriction
3. Epidemiological evidence from the JACC Study showing inverse associations between fat intake and gastric cancer risk
4. Temporal trends and international comparisons consistent with a protective role for dietary fat

The hypothesis provides a novel explanation for the Japanese diet paradox and suggests that the assumption "lower fat is always better" requires reconsideration for gastrointestinal health. Dietary fat optimization (20–30% of energy) should be considered as a complementary gastric cancer prevention strategy, particularly in populations with other risk factors.

Declaration of Generative AI in Scientific Writing

The author used AI-assisted tools for language editing, grammar checking, and style refinement during the preparation of this manuscript. The hypothesis, analysis, interpretation of literature, and conclusions are entirely the author's own.

Conflict of Interest

The author declares no conflicts of interest.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

[1] H. Sung, J. Ferlay, R. L. Siegel, et al., Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers

in 185 countries, CA: A Cancer Journal for Clinicians 71 (3) (2021) 209–249.

[2] D. C. Willcox, B. J. Willcox, H. Todoriki, M. Suzuki, The Okinawan diet: health implications of a low-calorie, nutrient-dense, antioxidant-rich dietary pattern low in glycemic load, Journal of the American College of Nutrition 28 (Suppl) (2009) 500S–516S.

[3] Ministry of Health, Labour and Welfare, Japan, National health and nutrition survey 2019 (2020).

[4] Chinese Nutrition Society, Chinese Dietary Guidelines (2022), People's Medical Publishing House, Beijing, 2022.

[5] L. D'Elia, G. Rossi, R. Ippolito, F. P. Cappuccio, P. Strazzullo, Habitual salt intake and risk of gastric cancer: a meta-analysis of prospective studies, Clinical Nutrition 31 (4) (2012) 489–498.

[6] N. Uemura, S. Okamoto, S. Yamamoto, et al., *Helicobacter pylori* infection and the development of gastric cancer, New England Journal of Medicine 345 (11) (2001) 784–789.

[7] T. Kamada, K. Haruma, M. Ito, et al., Time trends in *Helicobacter pylori* infection and atrophic gastritis over 40 years in Japan, Helicobacter 20 (3) (2015) 192–198.

[8] J. Powles, S. Fahimi, R. Micha, et al., Global, regional and national sodium intakes in 1990 and 2010: a systematic analysis of 24 h urinary sodium excretion and dietary surveys worldwide, BMJ Open 3 (12) (2013) e003733.

[9] C. Holcombe, *Helicobacter pylori*: the African enigma, Gut 33 (4) (1992) 429–431.

[10] M. Inoue, S. Tsugane, Epidemiology of gastric cancer in Japan, Postgraduate Medical Journal 81 (957) (2005) 419–424.

[11] J. R. Turner, Intestinal mucosal barrier function in health and disease, Nature Reviews Immunology 9 (11) (2009) 799–809.

[12] C. Papadia, G. Coruzzi, C. Montana, et al., Gastrointestinal mucosal injury and repair, in: L. R. Johnson (Ed.), Physiology of the Gastrointestinal Tract, 5th Edition, Academic Press, 2012, pp. 1087–1121.

[13] B. A. Hills, B. D. Butler, L. M. Lichtenberger, Gastric mucosal barrier: hydrophobic lining to the lumen of the stomach, American Journal of Physiology 244 (5) (1983) G561–568.

[14] L. M. Lichtenberger, The hydrophobic barrier properties of gastrointestinal mucus, Annual Review of Physiology 57 (1995) 565–583.

[15] I. A. Kirpich, W. Feng, Y. Wang, et al., The type of dietary fat modulates intestinal tight junction integrity, gut permeability, and hepatic toll-like receptor expression in a mouse model of alcoholic liver disease, Alcoholism: Clinical and Experimental Research 36 (5) (2012) 835–846.

[16] D. Hollander, A. Tarnawski, Is there a role for dietary essential fatty acids in gastroduodenal mucosal protection?, Journal of Clinical Gastroenterology 13 (Suppl 1) (1991) S72–74.

[17] J. L. Wallace, Prostaglandins, NSAIDs, and gastric mucosal protection: why doesn't the stomach digest itself?, Physiological Reviews 88 (4) (2008) 1547–1565.

[18] P. C. Calder, Functional roles of fatty acids and their effects on human health, JPEN Journal of Parenteral and Enteral Nutrition 39 (1 Suppl) (2015) 18S–32S.

[19] S. J. Konturek, T. Brzozowski, J. Majka, J. Pytko-Polonczyk, J. Stachura, Implications of essential fatty acid deficiency on stress ulcer and cytoprotection, Journal of Clinical Gastroenterology 13 (Suppl 1) (1991) S32–43.

[20] W. Schepp, B. Steffen, H.-J. Ruoff, V. Schusdziarra, M. Classen, Modulation of rat gastric mucosal prostaglandin E2 release by dietary linoleic acid: effects on gastric acid secretion and stress-induced mucosal damage, Gastroenterology 95 (1) (1988) 18–25.

[21] M. Wolters, J. Ahrens, M. Romaní-Pérez, et al., Dietary fat, the gut microbiota, and metabolic health – a systematic review conducted within the MyNewGut project, Clinical Nutrition 38 (6) (2019) 2504–2520.

[22] L. Peng, Z.-R. Li, R. S. Green, I. R. Holzman, J. Lin, Butyrate enhances the intestinal barrier by facilitating tight junction assembly via activation of AMP-activated protein kinase in Caco-2 cell monolayers, Journal of Nutrition 139 (9) (2009) 1619–1625.

[23] L. E. M. Willemsen, M. A. Koetsier, S. J. H. van Deventer, E. A. F. van Tol, Short chain fatty acids stimulate epithelial mucin 2 expression through differential effects on prostaglandin E1 and E2 production by intestinal myofibroblasts, Gut 52 (10) (2003) 1442–1447.

[24] S. Tsugane, M. Tsuda, F. Gey, S. Watanabe, Cross-sectional study with multiple measurements of biological markers for assessing stomach cancer risks at the population level, Environmental Health Perspectives 98 (1992) 207–210.

[25] N. Tokui, T. Yoshimura, Y. Fujino, et al., Dietary habits and stomach cancer risk in the JACC study, Journal of Epidemiology 15 (Suppl 2) (2005) S98–108.

[26] K. Katanoda, T. Matsuda, A. Matsuda, et al., An updated report of the trends in cancer incidence and mortality in Japan, Japanese Journal of Clinical Oncology 43 (5) (2013) 492–507.

[27] P. Bertuccio, L. Chatenoud, F. Levi, et al., Recent patterns in gastric cancer: a global overview, International Journal of Cancer 125 (3) (2009) 666–673.

[28] L. N. Kolonel, A. M. Y. Nomura, T. Hirohata, J. H. Hankin, M. W. Hinds, Association of diet and place of birth with stomach cancer incidence in Hawaii Japanese and Caucasians, American Journal of Clinical Nutrition 34 (11) (1981) 2478–2485.

[29] P. Correa, A human model of gastric carcinogenesis, Cancer Research 48 (13) (1988) 3554–3560.